

Admissible Non-Injective Transitions as the Primitive of Physical Description

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Abstract

Quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution. This paper establishes the minimal axiomatic foundation from which this claim follows: physical structure is derived from a single primitive, the local structure of admissible transitions between observable states. Four axioms govern this structure: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1 and A2 alone, we derive irreversibility, the arrow of time, and the existence of structural fluctuations, without any additional postulate. From A3, we derive the proto-state as a physically real unresolved configuration shared between two successive states, and show that admissibility forces the retention of phase coherence. We further prove that A3 forbids any admissible factorisation of the fibre, which forces a non-trivial commutator between conjugate generators; combined with minimality (A1) and Born–Infeld parity, this identifies the symmetry group of the admissible fibre as $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q , and the fibre itself as the Weil representation V_ρ — closing the identification $F_n \simeq V_\rho$ as a theorem rather than a remark. From A4, we derive the discrete character of quantum transitions as a consequence of the interaction between continuous Born–Infeld saturation and the discrete shell structure of the observable space. This paper provides the axiomatic foundation cited by the companion papers [8, 1, 13] and serves as the formal basis for the derivations carried out therein.

Keywords. Non-injective projection; Emergence; Projective structure; Admissibility; Born–Infeld constraint; Information loss; Projection entropy; Quantum emergence; Phase coherence; $\text{SU}(2)$ structure; Gauge emergence; Fiber structure; Projective temporal ordering; Irreversibility.

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1 Introduction

The Cosmochrony programme [4] aims to derive the structure of physical description from a minimal relational foundation, without postulating a background spacetime, a Hilbert space, or an external substrate. The only primitive is the local structure of admissible transitions between observable states.

The central question driving the framework is:

Given an observable state O_{n-1} , what states are accessible from it?

The answer to this question — encoded in the admissibility structure — is sufficient to derive irreversibility, temporal order, wave-like propagation, phase coherence, and the discrete character of quantum transitions. No substrate, no background, and no quantum postulate is required as input.

Relation to companion papers

This paper provides the axiomatic foundation for a chain of companion results. The companion paper [1] establishes that Bell-type factorisation fails generically under non-injective projection (A2), without invoking nonlocal dynamics. The companion paper [13] proves that non-injectivity is a structural necessity of any genuinely emergent description. The companion paper [8] derives phase coherence, the singlet correlator, the Tsirelson bound, and the Born rule as structural consequences of admissibility — using Axioms A1–A4 as stated here, combined with the concrete spectral results of the O-series. The white paper [4] develops the full phenomenological programme; the present axiomatic reformulation is a foundational companion to it, not yet integrated into its main text.

Organisation

Section 2 states the four axioms. Section 3 clarifies the role of Π and the index n . Section 4 derives the structural consequences of non-injectivity: irreversibility, novelty, temporal order, and the projective incompleteness principle. Section 5 derives the proto-state, phase, coherence, and the Heisenberg structure of the admissible fibre from the admissibility principle. Section 5.7 derives projection locking and the discrete character of quantum transitions. Section 6 treats entanglement and Bell non-applicability. Section 7 summarises what is derived from what, and identifies the open directions.

2 The Four Axioms

We state the four axioms from which the framework proceeds. They are not independent postulates chosen for convenience — each captures a structural feature that is either necessary for genuine emergence [13] or forced by the requirement of bounded effective descriptions [3].

Definition 2.1 (Admissible transition). *A transition from observable state O_{n-1} to observable state O_n is admissible if it satisfies Axioms A1–A4 below.*

Remark 2.2 (Admissibility is relational, not absolute). *Admissibility is never an intrinsic predicate of an observable state O_n taken in isolation. It is a relational constraint on projective transitions: a state O_n is admissible only relative to a preceding state O_{n-1} , whose accumulated constraints determine the local admissible neighbourhood. We write $O_n \in \mathcal{A}(O_{n-1})$ to make this explicit, where*

$$\mathcal{A}(O_{n-1}) := \{ O_n \mid \text{the transition } O_{n-1} \rightarrow O_n \text{ satisfies A1–A4} \} \quad (1)$$

is the admissible neighbourhood from O_{n-1} . The phrasing “ O_n is admissible” should always be read as shorthand for $O_n \in \mathcal{A}(O_{n-1})$: admissibility is always from-here admissibility, conditioned on the already-realized projective position. There is no fixed space of possible observable states: there is a field of local admissible neighbourhoods, each determined by the point already reached.

A1 — Local projective admissibility. For any observable state O_{n-1} , there exists a non-empty set F_n of admissible successor directions: configurations of O_{n-1} compatible with a transition to a stable next state. The set F_n is defined by internal structural constraints (Born–Infeld bound, spectral envelope, symmetry requirements) and does not presuppose an ambient configuration space.

A2 — Structural non-injectivity. Distinct admissible directions from O_{n-1} may lead to the same resolved successor state O_n . The map Π_n encoding this is generically non-injective.

A3 — Admissibility as non-premature selection. An admissible transition preserves the full multiplicity of open directions until projection locking occurs. It does not select among them prematurely — that is, before the Born–Infeld saturation threshold is reached [3].

A4 — Projection locking. Resolution occurs only when continuous Born–Infeld saturation meets the discrete shell support of the observable space. The transition from admissible to admitted is structurally discrete.

Remark 2.3 (Independence of the axioms). *A1 asserts existence of admissible directions. A2 asserts their generic non-injectivity. A3 constrains which transitions are admissible. A4 governs when resolution occurs. The four axioms are logically independent: none follows from the others. Their joint necessity is argued in Section 3 and in the companion papers [13, 3].*

Remark 2.4 (Status of Born–Infeld in A3 and A4). *The Born–Infeld saturation threshold appearing in A3 and A4 is not an additional postulate: it is the unique bounded completion of the effective action compatible with bounded projective flux, established in [3]. A3 and A4 therefore do not introduce new structure beyond A1–A2 and the Born–Infeld companion.*

Remark 2.5 (What projection locking selects). *A4 resolves the transition to a unique observable state O_n in the image of Π_n . It does not select a unique underlying configuration in F_n .*

The multiplicity of admissible configurations in the fibre $F_n = \Pi_n^{-1}(O_n)$ is not eliminated by projection locking: distinct configurations $f, f' \in F_n$ with $\Pi_n(f) = \Pi_n(f') = O_n$ remain structurally present after locking, but become observationally inaccessible. The non-injectivity of Π_n therefore persists at every resolution step.

This distinction — observable uniqueness without configurational uniqueness — prevents Bell-type factorisation [1] and forces the coherent, non-factorisable fibre structure derived in Section 5.4. More generally, distinct interactions $I_1 \neq I_2$ induce distinct admissible sub-fibres $F_n^{(I_1)} \neq F_n^{(I_2)}$ and hence distinct observable outcomes, without any element of F_n having been selected. This is the structural origin of contextuality: measurement context determines which sub-fibre is accessed, not which configuration is realised. This contextuality is synchronic: it concerns the choice of sub-fibre at a fixed transition step n . The admissible neighbourhood $\mathcal{A}(O_{n-1})$ (Remark 2.2) introduces a second, diachronic form of contextuality: the set of accessible observable states at step n is itself conditioned on the previously realized state O_{n-1} . The context of a projection is therefore not merely the measurement algebra chosen at step n , but also the already-realized projective position from which the next state may be reached. Standard contextuality frameworks (Kochen–Specker, sheaf-theoretic) capture synchronic contextuality; the present framework extends this to diachronic contextuality, in which the history of realized projections is constitutive of what is admissible next.

Remark 2.6 (Projective Observability Principle). *Axioms A1–A4 jointly imply the following criterion, which identifies what is physical in the framework.*

Physical observables coincide with the invariants of the projection Π_n : any quantity that depends on the choice of pre-image within $\Pi_n^{-1}(O_n)$ is unphysical by construction.

This is not an additional axiom but a consequence of A2 (which forces non-trivial fibres) and A4 (which identifies O_n with $\text{Im } \Pi_n$ exactly): since Π_n is the only structure that relates the configurational level to the observable level, any quantity invisible to Π_n is invisible to physics. What the projection erases — intra-fibre distinctions — is unphysical; what it preserves — the invariants of the fibre structure — constitutes the physical content of the theory. In particular, the symmetry group of a fibre $\Pi_n^{-1}(O_n)$ is physical precisely because it acts within the fibre while leaving O_n unchanged: this is the structural origin of gauge symmetry, developed in the companion papers [9, 27]. A detailed formulation in terms of the five-level hierarchy is given in Remark 3.5.

3 The Primitive and the Role of Π

3.1 The primitive

The only primitive of this framework is the pair (O_{n-1}, F_n) : an observable state and the set of admissible directions it admits. There is no external substrate, no hidden carrier, no background geometry. The fibre F_n is not defined in an abstract ambient space — it is induced by O_{n-1} itself through the admissibility constraints of A1–A3.

The resulting evolution is recursive: once O_n is realized, it immediately becomes the new departure point, generating the next admissible neighbourhood $\mathcal{A}(O_n)$ from which O_{n+1} may be reached:

$$O_{n-1} \longrightarrow O_n \longrightarrow \mathcal{A}(O_n) \longrightarrow O_{n+1}. \quad (2)$$

The realized state is simultaneously a destination and the generator of the next set of constraints. No external agent resets the context between steps: the admissible neighbourhood at each stage is determined entirely by the state just reached.

3.2 The projection Π_n as descriptive abstraction

The notation Π_n is a compact encoding of the local admissibility structure. It encodes three things simultaneously:

- (i) the admissible directions from O_{n-1} — the fibre F_n (A1, A2);
- (ii) the Born–Infeld saturation bound — the limit on accessible amplitudes (A3);
- (iii) projection locking — the constraint that resolution falls on an available shell (A4).

Π_n is not a physical entity and not a dynamical map. It is a structural relation encoding the admissible identification between distinguishable configurations (elements of F_n) and indistinguishable observable outcomes (elements of O_n). Saying “ Π_n is non-injective” means that multiple directions from O_{n-1} can lead to the same O_n — a fact about the admissible transition structure, not a property of an external map. Non-injectivity does not originate from an underlying substrate, but from the structure of admissible configurations and their interaction-dependent realisation map.

Remark 3.1 (The threefold role of Π_n). *The structural consequences established below in Sections 4 and 4.4 reveal that Π_n plays three distinct roles simultaneously, each irreducible to the others.*

- (i) Admissibility filter. Π_n selects, from the structurally compatible continuations $\Omega_n^{(c)}$, those directions satisfying the Born–Infeld and spectral constraints of A1–A3. What it retains constitutes the observable O_n ; what it discards is the non-trivial kernel whose existence is guaranteed by Corollary 4.6.
- (ii) Generator of temporal order. The oriented sequence $\Pi_0, \Pi_1, \Pi_2, \dots$ carries the partial order established in Theorem 4.2. Each Π_n is a node in the admissibility graph (Remark 4.4); the index n acquires its temporal meaning from this orientation, not from an external parameter. The dynamics of the framework is the oriented topology of this graph — not a law superimposed on a pre-existing time.
- (iii) Revelation, not creation. Π_n does not produce new structure in χ . It selects, from the admissible spectral modes of the Weil representation already present in $\Omega_n^{(c)}$, those that are observationally accessible at depth n . The BFS depth controls how much of the structure of χ has been revealed; increasing n does not modify χ but resolves finer spectral content — as a deeper lens resolves detail that was always present in the object, not detail that the lens creates.

These three roles are inseparable: the structural constraints A1–A3 and Born–Infeld parity that force the admissibility filter simultaneously generate the temporal order via Theorem 4.2 and guarantee the incompleteness of every resolution via Corollary 4.6. The non-commutativity $[X, \sigma(X)] \neq 0$ at the root of the incompleteness (Theorem 5.7) is therefore not merely a technical property of the fibre: it is the algebraic source of all three roles at once.

3.3 The index n

The index n does not represent a pre-existing time parameter. It counts transitions: the number of times a state has been succeeded by another under the admissibility constraints. Its temporal meaning is derived in Section 4 — it is not assumed.

3.4 Where non-injectivity lives

A natural question arises: where does the non-injectivity of Π_n come from, given that no underlying substrate is postulated? The answer is precise.

Non-injectivity is a property of the interaction-dependent realisation map: the map from the space F_n of structurally compatible configurations to the space O_n of observable outcomes. When two distinct admissible configurations $f, f' \in F_n$ give rise to the same observable outcome $o \in O_n$, it is not because they are “the same at a deeper level” — it is because the interaction

that realises the transition cannot distinguish them within O_n . The limit on distinguishability is set by the Born–Infeld capacity constraint (A3): the observable space O_n has finite projective capacity, and configurations that differ only below this threshold are identified.

Three properties of this realisation map are essential:

- (i) *Interaction-dependence*: Π_n encodes the interaction; different interactions define different realisation maps and different fibres F_n .
- (ii) *Internal definition of admissibles*: F_n is defined by internal constraints (BI bound, spectral envelope, symmetry) — not by reference to an ambient substrate.
- (iii) *Minimal structure of F_n* : the admissibility constraints force F_n to carry a specific algebraic structure — $SU(2)$ in the spin- $\frac{1}{2}$ sector (O23) — which is derived from the constraints, not assumed.

The non-injectivity is therefore neither a property of a hidden substrate nor an arbitrary assumption: it is a structural consequence of finite projective capacity applied to an internally defined space of admissible configurations.

Remark 3.2 (The admissible fibre as equivalence class under projection). *A useful image for the non-injectivity of Π_n is that of shadows cast by a fixed projection setup. A given shadow does not correspond to a single object: it corresponds to the set of all objects whose projection yields that shadow — the admissible fibre $F_n = \Pi_n^{-1}(O_n)$.*

The directionality is essential. The fibre is not the collection of all shadows produced by a given object under varying projections (one configuration, many images). It is the collection of all configurations that produce the same observable under a fixed, non-injective Π_n (many configurations, one observable). This many-to-one structure is the source of irreversibility (Section 4) and of the non-factorisability of the proto-state (Section 5.4).

Remark 3.3 (Connection to the O-series: F_n and V_ρ). *In the concrete spectral realisation of the O-series, the admissible fibre F_n is realised as a spectral sector V_ρ of the Heisenberg Cayley graph $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$. The sectors V_ρ are not pre-existing configuration spaces: they are the structures induced by the admissibility constraints — the Born–Infeld bound, the spectral envelope, and the symmetry requirements of A1. In this sense, V_ρ is the concrete realisation of F_n , not an ambient space in which F_n is embedded:*

$$F_n \simeq V_\rho. \tag{3}$$

This identification, stated here as a structural connection to the O-series, is established as a theorem in Section 5.4: the axioms A1–A3 and Born–Infeld parity together force the symmetry group of F_n to be $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and the Stone–von Neumann theorem then identifies $F_n \simeq V_\rho$ uniquely.

Remark 3.4 (The two levels of Π_n : kernel and image (O24)). *The non-injective projection Π_n operates at two structurally distinct levels, established in O24 [23]:*

- (i) Kernel level ($\ker \Pi_n$): *governs microscopic multiplicity and fluctuation amplitude. Multiple configurations in $\Omega_n^{(c)}$ may project to the same observable outcome; their differences are invisible to admissible observables and appear as projection residuals. The dependence on O_{n-1} — via the baseline $V_{n-1}^{(c)}$ — enters at this level: it determines which directions in $\Omega_n^{(c)}$ are new relative to the already realized subspace.*
- (ii) Image level ($\text{Im } \Pi_n \cap \text{Ntrl}$): *governs the observable structure. Its real rank is rigidly fixed to 3 by the quaternionic maximality of $\text{Im } \mathbb{H}$ (O23, Theorem 3.2) and is independent of the cardinality of $\ker \Pi_n$.*

The Verticality Lemma of O24 (Lemma 3.1) establishes that every Born–Infeld-admissible symmetry acts vertically: it maps each fibre $\Pi_n^{-1}(o)$ to itself and does not increase the rank of the image. The structural consequence is:

Non-injectivity affects degeneracy, not structure.

Enlarging $\ker \Pi_n$ — admitting more configurations in each fibre — increases the amplitude of quantum fluctuations but does not alter the observable rank, the cascade exponent δ_{pair} , or any other structural invariant of the admissible sector. The two levels of Π_n are therefore decoupled: kernel structure is physically relevant for fluctuations; image structure is physically relevant for observables.

Remark 3.5 (The observable space principle: $\mathcal{O} = \text{Im } \Pi$). *A direct consequence of the five-level hierarchy is that admissible observables are functions on*

$$\mathcal{O} := \text{Im } \Pi = \bigsqcup_n O_n, \quad (4)$$

not on the pre-image space $\Omega_n^{(c)}$. Any structural difference between two configurations $f, f' \in \Pi_n^{-1}(O_n)$ that share the same image under Π_n is invisible to every admissible observable by definition: it is a difference within the fibre, which the projection identifies as observationally indistinguishable. This is not an additional assumption but an immediate consequence of Definition 3.6 and Level 5: O_n is defined as $\text{Im } \Pi_n$, and admissibility is a property of Π_n acting on $\Omega_n^{(c)}$, not of $\Omega_n^{(c)}$ itself. Intra-fibre variability is therefore structurally unobservable; it constitutes the degree of freedom whose projected manifestation appears as quantum indeterminacy in the effective description. In particular, any admissible quantity that depends on the choice of representative in $\Pi_n^{-1}(O_n)$ must agree for all representatives in the same fibre class: this is the structural basis for the uniformity of admissible spectral invariants across the fibre.

The abstract structure of Section 3 has a concrete realisation in the O -series that answers precisely where Ω_n lives and how it is constructed. The realisation proceeds through five levels, each induced by the previous one.

Level 1 — The relational group G_q . The combinatorial substrate of the BFS exploration is the discrete Heisenberg group

$$G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), \quad |G_q| = q^3, \quad (5)$$

equipped with the standard symmetric generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. G_q is not a physical space: it is a relational structure, encoding how configurations relate to one another via the generators. The BFS explores G_q in shells $S_n = B_n \setminus B_{n-1}$, where $B_n = \{g \in G_q : d(e, g) \leq n\}$. The shell structure $\{S_n\}$ is the concrete realisation of the transition-counting index n of Section 3.

Level 2 — The Weil representation ρ_c and its ambient space. For each non-zero character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, the discrete Heisenberg group G_q admits an irreducible q -dimensional Weil representation [6]:

$$\rho_c : G_q \longrightarrow \text{GL}(\mathbb{C}^q), \quad (\rho_c(a, b, \gamma) f)(x) = e^{2\pi i c(\gamma + bx)/q} f(x - a), \quad (6)$$

acting on $L^2(\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{C}^q$. The ambient space $\mathbb{C}^q$ is not a physical configuration space presupposed by the theory: it is the natural carrier of the irreducible representation of G_q of central character c . It is induced by G_q and c , not by any external assumption.

Level 3 — The orbit span Ω_n as emerging subspace. Given the initial vector $v_0 \in \mathbb{C}^q$ and the Weil action ρ_c , the orbit span at BFS depth n is:

$$\Omega_n^{(c)} := \text{span}\{\rho_c(g) v_0 : g \in B_n\} \subseteq \mathbb{C}^q. \quad (7)$$

By construction, $\Omega_n^{(c)}$ is a minimal extension of the previously realized subspace:

$$V_{n-1}^{(c)} \subseteq \Omega_n^{(c)} \subseteq \mathbb{C}^q, \quad (8)$$

where $V_{n-1}^{(c)} = \text{GS}(B_{n-1})$ is the Gram–Schmidt span accumulated over the preceding shells. $\Omega_n^{(c)}$ is not defined by O_{n-1} , but is constrained by the already realized subspace $V_{n-1}^{(c)}$: the directions in $\Omega_n^{(c)} \setminus V_{n-1}^{(c)}$ are the candidates for new observable structure at depth n .

Definition 3.6 (Structurally compatible continuations). $\Omega_n^{(c)}$ is the space of structurally compatible continuations from O_{n-1} : it contains every direction reachable by the relational structure of G_q via ρ_c , without any admissibility filter applied. Admissibility is not a property of $\Omega_n^{(c)}$; it is a property of Π_n acting on $\Omega_n^{(c)}$.

Three properties follow immediately:

- (i) *Emergence*: $\Omega_n^{(c)}$ is not given a priori — it is built incrementally as the BFS explores G_q .
- (ii) *Indirect conditioning on O_{n-1}* : $\Omega_n^{(c)}$ is defined by the group action alone — the definition (7) refers only to B_n and ρ_c . The dependence on O_{n-1} is indirect: O_{n-1} determines the Gram–Schmidt span GS_{n-1} already accumulated over B_{n-1} , which acts as the baseline against which new directions in $\Omega_n^{(c)}$ are measured. This baseline is used by Π_n (not by $\Omega_n^{(c)}$ itself) to identify genuinely new directions, counted by $\sigma_c(n) = \delta r_n / |S_n|$ [6]. In summary: $\Omega_n^{(c)}$ is determined by (G_q, ρ_c, B_n) ; the role of O_{n-1} enters only through Π_n .
- (iii) *Pre-admissibility character*: $\Omega_n^{(c)}$ contains all directions generated by the group action of G_q up to depth n — including directions that Π_n will subsequently reject as inadmissible. The admissibility filter is entirely in Π_n , not in $\Omega_n^{(c)}$.

Level 4 — The admissibility filter Π_n . The projection Π_n acts on $\Omega_n^{(c)}$ via the BI admissibility constraint $A_n \leq c_\chi / \sqrt{\lambda_n}$ and the Gram–Schmidt orthogonalisation [6, 18]. It selects, from all directions in $\Omega_n^{(c)}$, those compatible with the admissibility constraints. Π_n encodes the laws revealed by the O-series — Born–Infeld bound, parity involution, projection locking — but does not define $\Omega_n^{(c)}$.

Level 5 — The observable outcome O_n . The observable state O_n is the image of Π_n applied to the admissible part of $\Omega_n^{(c)}$. It is a discrete, shell-indexed element of the observable space [18]:

$$O = \bigsqcup_{n \in \mathbb{N}_q} O_n. \quad (9)$$

Remark 3.7 (Summary: the five-level hierarchy). *The five levels constitute a single construction with no external inputs:*

$$G_q \xrightarrow{\rho_c} \mathbb{C}^q \xrightarrow{\text{BFS}} \Omega_n^{(c)} \xrightarrow{\Pi_n} O_n, \quad V_{n-1}^{(c)} \subseteq \Omega_n^{(c)} \subseteq \mathbb{C}^q. \quad (10)$$

G_q is the relational structure. $\mathbb{C}^q$ is its representation carrier. $\Omega_n^{(c)}$ is the emergent space of structurally compatible continuations (not yet filtered by admissibility), constrained from below by the already realized subspace $V_{n-1}^{(c)}$. Π_n is the admissibility filter. O_n is the resolved observable state. No level presupposes the next: each is induced by the preceding structure and the admissibility constraints. This is the concrete answer to the question of where

Ω_n lives: in $\mathbb{C}^q$, itself induced by G_q — not in an external configuration space.

Remark 3.8 (The conjugate pair and Ω_n without preferred direction). The parity involution $c \leftrightarrow q - c$ (O18) acts on $\Omega_n^{(c)}$ via complex conjugation [7]: $\Omega_n^{(q-c)} = \overline{\Omega_n^{(c)}}$. The minimal admissible fibre is therefore the pair $\{V_n^{(c)}, V_n^{(q-c)}\} = \{V_n^{(c)}, \overline{V_n^{(c)}}\}$, with no preferred orientation — consistent with the admissibility principle of A3.

3.5 From $\mathbb{C}^q$ to continuous spacetime: the large- q limit

The five-level hierarchy of Section 3.4 is formulated for finite prime q . Physical spacetime is continuous. This subsection identifies the structural bridge between the two, without claiming to complete the derivation.

The discrete structure at finite q . At finite q , the observable space $O = \bigsqcup_{n \in \mathbb{N}_q} O_n$ is discrete and finite. The ambient representation space $\mathbb{C}^q$ has dimension q . Physical quantities are functions on $\mathbb{Z}/q\mathbb{Z}$.

The large- q limit. As $q \rightarrow \infty$ with the prime structure retained, three things happen simultaneously:

- (i) *The representation space $\mathbb{C}^q$ approaches $L^2(\mathbb{R})$.* Functions on $\mathbb{Z}/q\mathbb{Z}$ with fixed Fourier support become functions on $\mathbb{R}$ in the Pontryagin dual limit. The Weil representation ρ_c of G_q approaches the metaplectic representation of the real Heisenberg group $\text{Heis}_3(\mathbb{R})$.
- (ii) *The BFS shell structure approaches a continuous foliation.* The polynomial ball growth $|B_n| \sim cn^4$ (Bass–Guivarc’h, homogeneous dimension $D = 4$) induces, in the $q \rightarrow \infty$ limit, a continuous layering of the configuration space compatible with a four-dimensional effective geometry [4].
- (iii) *The discrete admissibility filter Π_n approaches a differential constraint.* The Born–Infeld admissibility bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ becomes, in the continuum limit, the Born–Infeld Lagrangian constraint on effective field configurations [3].

The continuum spacetime observable is therefore not a separate postulate: it is the large- q limit of the discrete observable space O , with the four-dimensional structure inherited from the homogeneous dimension of G_q .

Remark 3.9 (Status of the continuum limit). The convergence $\mathbb{C}^q \rightarrow L^2(\mathbb{R})$ and the emergence of continuous spacetime geometry from the BFS shell structure are established qualitatively in [4]. The companion paper [11] initiates the rigorous treatment of this limit, formulating it in the category of

Hilbert spaces and operator algebras and decomposing it into three distinct convergences: a Hilbert inductive limit (T1), a representational convergence of Weil generators to metaplectic generators (T2), and a Mosco convergence of admissibility forms to a second-order operator L_Π on $L^2(\mathbb{R})$ (T3). The continuum limit is reduced to a single spectral tightness condition (Conjecture C of [11]), which is confirmed quantitatively across all tested primes: the spectral tail mass of admissible test vectors satisfies $E_q(q/3)/\|f_q\|^2 < 2.1 \times 10^{-5}$ uniformly, with values decreasing toward machine precision as q grows. The embedding hypothesis [H1] is proved in the low-frequency regime relevant to admissible sequences; the Weil-block compatibility condition and the identification of the effective metric from the symbol of L_Π are established in the companion paper Q5b [2]. The present remark identifies the mechanism; [11] establishes the operator-theoretic framework and the numerical evidence.

Remark 3.10 (Why four dimensions). *The homogeneous dimension $D = 4$ of the Heisenberg group G_q (Bass–Guivarc’h theorem: $|B_n| \sim n^4$) is not chosen to match the observed four dimensions of spacetime. It is a structural consequence of the nilpotency class and rank of Heis_3 : the group has three generators with one non-trivial commutator, giving $D = 2 \cdot \text{rank} + \text{class} = 4$. The coincidence with the observed dimension of spacetime is therefore a structural prediction of the framework, not a parameter fit.*

4 Structural Consequences of Non-Injectivity

Non-injectivity (A2) and the existence of distinct successor states together imply three structural facts whose sources are distinct.

4.1 Irreversibility

Proposition 4.1 (Irreversibility from non-injectivity). *Under A2, every admissible transition involves an irreversible loss of structural information.*

Proof. Since multiple directions from O_{n-1} map to the same O_n under Π_n , the information distinguishing these directions is absent from O_n . No subsequent transition $\Pi_{n'}$ for $n' > n$ can recover this information: the map Π_n has positive projection entropy $S_{\Pi_n} > 0$ [13], and no admissible transition can reduce S_{Π} below zero. The loss is therefore structural and permanent. $\square$

4.2 Novelty and structural fluctuations

The existence of a distinct successor state $O_n \neq O_{n-1}$ defines a structural residue:

$$\Delta_n := \text{Res}(O_{n-1}, O_n; \Pi_n), \quad (11)$$

the measure of how much O_n is irreducible to a continuation of O_{n-1} under Π_n . That Δ_n is non-trivial is not an additional assumption — it is the condition under which we index O_n as a distinct state. These residues are what physical descriptions call fluctuations or quantum noise.

4.3 Temporal order

Theorem 4.2 (Temporal order from A2). *Under A2, the chain of observable states $\{O_n\}_{n \geq 0}$ carries a canonical oriented, non-trivial partial order.*

Proof. Irreversibility (Proposition 4.1) establishes an asymmetric relation: $O_{n-1} \rightarrow O_n$ is not reversible, so O_{n-1} and O_n are not equivalent under the transition structure. Novelty ($\Delta_n \neq 0$) ensures $O_n \neq O_{n-1}$, so the order is non-trivial. The composition of transitions defines transitivity. Together: asymmetry, irreflexivity, and transitivity give a strict partial order compatible with the chain structure. $\square$

Remark 4.3 (Time as derived structure). *The arrow of time is not postulated. It is the oriented order generated by the chain of non-injective transitions under A2. The index n acquires its temporal meaning from this order — not the reverse. The quantitative structure of this projective temporal ordering — cumulative projected capacity and its numerical validation from the spectral admissibility cascade — is developed in [14].*

Remark 4.4 (Admissible transitions as a state machine). *The admissible neighbourhood $\mathcal{A}(O_{n-1})$ (Remark 2.2) gives the chain (2) the structure of an oriented state machine: nodes are admissible projections, edges are admissible transitions, and a physical history is a directed path in this graph. When $|\mathcal{A}(O_{n-1})| = 1$, the successor is uniquely determined by the admissibility structure alone — no external law is required. When $|\mathcal{A}(O_{n-1})| > 1$, branching occurs; the relative admissibility weights of the branches are the projective analogues of quantum probabilities (Section 5). In both cases, no temporal parameter external to the transition structure is needed: the index n acquires its temporal meaning entirely from the orientation of the admissibility graph. Dynamics is the oriented topology of the admissibility graph, not a law imposed upon a pre-existing time.*

Remark 4.5 (Observable reality as projective position). *The admissible neighbourhood $\mathcal{A}(O_{n-1})$ is not a fixed space of possibilities. It shifts at every step, because O_n conditions $\mathcal{A}(O_n)$, which differs generically from $\mathcal{A}(O_{n-1})$: the projection following step n is never performed from an invariant reference frame. Observable reality is therefore not a static projection of a fixed substrate onto a fixed observable space. It is the current projective position within the space of admissible reprojections: the trajectory $O_0 \rightsquigarrow O_1 \rightsquigarrow \dots \rightsquigarrow O_n$ is constitutive of what is real at step n , not contingent upon it. The cumulative index $I(n)$ measures not elapsed time but accumulated projective divergence from the initial position — how far the sequence has moved through the space of admissible neighbourhoods. In particular, the projection does not reveal a pre-existing reality: it redefines the context from which the next reality may appear.*

4.4 Projective incompleteness

Theorem 5.7 establishes that the admissibility structure forces $[X, \sigma(X)] = Z \neq 0$. This non-commutativity has an immediate structural consequence that is logically prior to — and independent of — the wave and phase structure of Section 5.

Corollary 4.6 (Projective incompleteness). *Under A1–A3 and Born–Infeld parity, no admissible projection Π_n resolves $\Omega_n^{(c)}$ completely. Every admissible projection has a non-trivial kernel and leaves a strictly positive informational residue:*

$$\ker \Pi_n \neq \{0\}, \quad S_{\Pi_n} > 0, \quad (12)$$

for every admissible Π_n .

Proof. By Theorem 5.7, $[X, \sigma(X)] = Z \neq 0$. A projection Π_n with $\ker \Pi_n = \{0\}$ would simultaneously resolve both X and $\sigma(X)$ without residual ambiguity, requiring their simultaneous diagonalisation and hence $[X, \sigma(X)] = 0$.

This contradicts $Z \neq 0$. Therefore $\ker \Pi_n \neq \{0\}$, and the projection entropy $S_{\Pi_n} > 0$ follows from Proposition 4.1. $\square$

Remark 4.7 (Structural fluctuations as necessary, not contingent). *Corollary 4.6 changes the epistemic status of the structural fluctuations introduced in Section 4.2. They are not merely a consequence of abstract non-injectivity (A2 alone), but a necessary consequence of the non-commutativity forced by A1–A3: any projection that left no residue would require $[X, \sigma(X)] = 0$, contradicting Theorem 5.7. The quantum vacuum is therefore not a state from which all fluctuations have been removed; it is the state from which complete removal is structurally forbidden.*

Remark 4.8 (Projective incompleteness and the first transition). *Corollary 4.6 also resolves a potential question about the origin of the first admissible transition: what initiates Π_1 from Π_0 ? A perfectly self-closed initial state — one with $\ker \Pi_0 = \{0\}$ — is structurally forbidden. The initial projection necessarily leaves a non-zero residue, which constitutes the first difference and guarantees that $\mathcal{A}(O_0)$ is non-empty. No external mechanism is required: the first transition is a structural necessity, not a contingent event. This is the projective analogue of the impossibility of a perfectly symmetric initial state: the admissibility constraints that select the Heisenberg structure also ensure that a featureless origin is inadmissible.*

5 Admissibility, Wave, and Phase Coherence

5.1 The admissibility principle

Axiom A3 introduces a physical constraint on transitions:

A transition is admissible if and only if it preserves the full multiplicity of open admissible directions without selecting among them prematurely.

Premature selection means making a determinative choice that nothing in the current state O_{n-1} justifies. Admissibility forbids it: the transition must remain structurally neutral among all open directions until projection locking (A4) forces resolution.

5.2 The proto-state

Definition 5.1 (Proto-state). *The proto-state Ψ_n associated with a transition from O_{n-1} to O_n is the unresolved configuration shared between O_{n-1} and O_n : the carrier of the transition, belonging fully to neither endpoint.*

The proto-state is physically real — it is the configuration of the system during the transition, before projection locking. It is not an epistemic representation of uncertainty about a definite state: it is the actual structural state of the system between two realized observable states.

Proposition 5.2 (Structural indeterminacy of the proto-state). *Under A1 and A3, the proto-state Ψ_n is genuinely indeterminate among the admissible directions in F_n .*

Proof. By A1, F_n is non-empty. By A3, the transition does not select prematurely among the elements of F_n . Therefore Ψ_n is not identified with any particular element of F_n before projection locking: it is open across all of F_n simultaneously. This is structural indeterminacy, not epistemic ignorance. $\square$

5.3 Wave, phase, and coherence

The proto-state, being structurally open across all elements of F_n simultaneously, propagates through all admissible directions at once. This is the physical origin of wave-like behaviour: not a metaphor, but a direct consequence of admissibility.

A configuration propagating through multiple directions simultaneously carries relative phase structure between those directions. Phase is therefore not postulated — it is a consequence of structural indeterminacy.

Admissibility (A3) preserves this phase structure: premature selection would collapse the relative phases, but A3 forbids it. The chain of consequences is therefore:

$$\begin{aligned}
 \text{A3} &\implies \text{structural indeterminacy} \\
 &\implies \text{proto-state (wave)} \\
 &\implies \text{phase} \\
 &\implies \text{coherence preserved.}
 \end{aligned} \tag{13}$$

Remark 5.3 (The admissible fibre as precursor of Hilbert structure). *The fibre F_n , equipped with the phase structure forced by A3, is the natural precursor of an effective Hilbert sector: it carries complex amplitudes and interference structure. This is not yet a full Hilbert space — the inner product, completeness, and probability measure require additional derivation (see Q1, Q2 in Section 7). But the amplitude structure and interference follow from A3 alone.*

Remark 5.4 (Double-slit experiment). *In the double-slit experiment, the proto-state propagates through both slits because the admissibility constraints do not select between them: both are open directions in F_n . The interference pattern is a direct consequence of coherent propagation through all admissible*

directions. The impact on the screen is the moment the proto-state reaches the Born–Infeld saturation threshold — projection locking (A4) forces it onto an available shell, resolving it into a localized state O_n .

5.4 Non-factorisability and the emergence of Heisenberg structure

Axiom A3 does not only enforce phase coherence. It also constrains the algebraic structure of the admissible fibre F_n itself, forcing it to carry a Heisenberg structure rather than an arbitrary unitary symmetry.

Remark 5.5 (The structural tension between A2 and A4). *The non-factorisability established in this section can be understood as the minimal consistent resolution of a structural tension between A2 and A4.*

A2 asserts that the multiplicity of admissible configurations in F_n cannot be eliminated: distinct configurations lead to the same observable, and this indistinguishability is structural, not epistemic. A4 asserts that the observable outcome is unique at locking. Together, they impose an irresolvable constraint: the multiplicity cannot disappear (A2 would be violated), yet it cannot remain directly observable (A4 would be violated).

A3 — admissibility as non-premature selection — is the operative axiom that determines how this tension resolves. It forbids any factorisation of F_n that would make the multiplicity distinguishable prior to locking, forcing it to organise into a globally coherent, irreducible structure. Lemma 5.6 formalises this constraint; Theorem 5.7 identifies the resulting structure as Heisenberg type. Quantum structure is therefore the minimal algebraic form compatible with the joint constraint of A2, A3, and A4.

Lemma 5.6 (Admissible non-factorisability). *Under A1–A3, any decomposition*

$$F_n = F^{(1)} \oplus F^{(2)}$$

that is stable under the admissible generators of F_n is forbidden prior to projection locking. In particular, A3 forces the action of admissible generators on F_n to be irreducible.

Proof. A3 forbids premature selection among open directions. A decomposition of F_n stable under admissible generators separates directions by invariants that are internal to the fibre — invariants carried by the generators themselves, which are constitutive of the admissibility structure by A1. Such a decomposition therefore resolves the proto-state partially, by assigning each open direction to a definite sector, prior to any Born–Infeld saturation event. This contradicts A3. Irreducibility follows immediately: any proper stable sub-representation would yield such a forbidden decomposition. $\square$

Theorem 5.7 (A1–A3 and BI parity force Heisenberg structure). *Under A1–A3 and Born–Infeld parity, the symmetry group of the admissible fibre F_n is isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q . Consequently, F_n is isomorphic to the Weil representation V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$.*

Proof sketch. Let X and $\sigma(X)$ be the conjugate generators of F_n under Born–Infeld parity. If $[X, \sigma(X)] = 0$, then F_n decomposes into simultaneous eigenspaces, yielding a stable decomposition forbidden by Lemma 5.6. Hence $[X, \sigma(X)] \neq 0$. Minimality (A1) forces the commutator $Z := [X, \sigma(X)]$ to be central: any non-central Z would extend the algebra beyond the conjugate pair, contradicting the minimal generative closure forced by A1. The group $\langle X, \sigma(X) \rangle$ is then a non-trivial central extension of $(\mathbb{Z}/q\mathbb{Z})^2$ by $\langle Z \rangle$, isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ by the classification of such extensions. Lemma 5.6 further implies that $|Z|$ is prime: if $|Z|$ were composite, the representation of F_n would decompose over the characters of $\langle Z \rangle$, yielding a forbidden stable decomposition. Finally, Born–Infeld parity forces $c \neq 0$ [3], so the central character is non-trivial; the discrete Stone–von Neumann theorem then identifies $F_n \simeq V_\rho$ uniquely. The detailed proof of each step is given in the companion paper [12]. $\square$

$$\begin{aligned}
\text{A3} &\implies \text{no admissible factorisation} \implies [X, \sigma(X)] \neq 0 \implies \text{central extension,} \\
\text{A1} &\implies \text{minimal generative closure} \implies \text{Heisenberg algebra,} \\
\text{A3} &\implies \text{irreducibility} \implies |Z| = q \text{ prime} \implies \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), \\
\text{BI parity} &\implies c \neq 0 \implies F_n \simeq V_\rho.
\end{aligned} \tag{14}$$

Remark 5.8 (Uncertainty principle as a corollary). *Theorem 5.7 subsumes the open direction Q6 of Section 7. The incompatibility of conjugate observables X and $\sigma(X)$ — i.e. the Heisenberg uncertainty principle — is a direct corollary: it follows from $[X, \sigma(X)] \neq 0$ together with the standard operator inequality. The uncertainty principle is therefore not a property of the observables, but a property of the non-resolved proto-state: it is the algebraic signature of the prohibition on premature selection encoded in A3.*

Remark 5.9 (Canonical quantisation as structural necessity). *The relation $[X, \sigma(X)] = Z$ defines a minimal discrete symplectic structure, of which the Heisenberg algebra is the canonical quantisation. Theorem 5.7 therefore establishes that canonical quantisation is not a modelling choice: it is the unique algebraic structure compatible with the requirement that the proto-state remain genuinely non-resolved until projection locking.*

A quantum superposition is an unresolved proto-state: a physically real configuration whose fibre maintains coherent relations between multiple

open admissible directions. There is no ontological paradox — superposition is the normal condition of a system between two transitions.

A measurement is a transition whose admissibility constraints are strong enough to force resolution of the proto-state onto a single admissible direction. Free evolution and measurement differ in the strength of their admissibility constraints, not in their fundamental mechanism.

In this sense, quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution.

5.5 The wavefunction as effective representation of the proto-state

A quantum wave is not a physical substance propagating in space. It is the effective representation of a physically real unresolved proto-state. It encodes the coherent structure of admissible directions prior to projection locking.

More precisely: the proto-state Ψ_n is real; the wavefunction ψ is a coordinate description of the coherent admissible structure carried by Ψ_n in the spectral realisation V_ρ of F_n . When we write ψ , we are not describing a substance propagating in space — we are encoding the relational phase structure of an unresolved configuration in the space of admissible directions. Quantum mechanics does not describe realized physical states: it describes states in the process of being resolved.

This resolves the standard interpretational difficulties without additional postulate:

- *Superposition* is the normal condition of a proto-state — the simultaneous structural openness across all elements of F_n .
- *Interference* is the direct consequence of coherent propagation through multiple admissible directions before resolution.
- *Collapse* is not a physical process applied to a pre-existing state but the resolution of the proto-state at projection locking — the transition from unresolved admissible structure to a realized observable outcome.

5.6 Why complex amplitudes

Given that the proto-state is a physically real, unresolved, coherent configuration, what is the minimal mathematical structure capable of representing it?

Real positive weights are insufficient. They describe ignorance about which of several pre-existing alternatives is realized — a classical mixture. The proto-state is not a mixture: it is a single coherent configuration with definite phase relations between admissible directions.

Real signed weights are also insufficient. They allow opposition of sign but do not support a rich enough phase structure under composition. In particular, they cannot represent the full coherence of a proto-state whose directions combine under the $SU(2)$ structure forced by O23.

The minimal structure that can: (i) represent simultaneous coherent openness across multiple directions, (ii) encode their relative phase relations, (iii) compose correctly under the $SU(2)$ symmetry of the admissible fibre, (iv) support interference — the physical consequence of coherence before resolution — is a complex amplitude.

Therefore: complex amplitudes are not introduced by convention. They are the minimal representation of a coherent unresolved proto-state. The imaginary unit encodes the relative phase between admissible directions; it enters because the proto-state is not a mixture but a coherent superposition.

5.7 Why the Born weight is the squared norm

Upon projection locking (A4), the unresolved proto-state is resolved onto a single admissible direction. The quantity associated with each direction at resolution — the *intensity of realisation* — must satisfy four structural requirements:

- (i) *Positivity*: a weight on admissible directions cannot be negative.
- (ii) *Global phase invariance*: multiplying ψ by a global phase $e^{i\theta}$ does not change the physical proto-state, so the weight must be invariant under this transformation.
- (iii) *Normalisability*: the total weight over all admissible directions must be finite and summable to one.
- (iv) *Coherence preservation before resolution*: the weight must not erase phase relations prematurely — interference must remain possible up to the moment of resolution.

The squared norm $|\psi|^2$ is the unique minimal quantity satisfying all four requirements. The norm $|\psi|$ fails (ii) under phase shifts and (iv) — it introduces real-valued interference which is not the correct structure. Any higher power $|\psi|^p$ for $p \neq 2$ fails (iii) under composition of superpositions. Any non-analytic function of ψ fails (iv).

The Born weight is therefore not a postulate. It is the positive intensity associated with the resolution of a coherent admissible structure — the unique minimal such intensity compatible with the four structural requirements.

Remark 5.10 (Relation to the Born rule derivation in the companion paper). *The argument above establishes the structural necessity of the squared*

norm as the resolution intensity of the proto-state. The companion paper [8] makes this precise as a theorem in the $SU(2)$ parity sector: the Born rule $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$ is the unique probability assignment compatible with the coherent fibre structure, the $SU(2)$ observable algebra, and the derived singlet correlator. The two derivations are complementary: the present argument gives the structural motivation; the companion gives the uniqueness proof.

Remark 5.11 (The wavefunction lives in the admissible fibre, not in space). A common source of confusion is to picture the wavefunction as an object propagating in physical space. In the present framework, ψ encodes the structure of F_n — the admissible directions from O_{n-1} . Its concrete realisation is the spectral sector V_ρ induced by the admissibility constraints on the Heisenberg Cayley graph (O -series). F_n and V_ρ are not pre-existing spaces: they are the spaces defined by the admissibility constraints. The wavefunction does not propagate in space; it encodes the relational phase structure of an unresolved admissible configuration.

Axiom A4 establishes that resolution is discrete. This discreteness is not postulated: it follows from the interaction of two structures already present in the framework.

5.8 Two types of discreteness

The Born–Infeld saturation condition [3] operates on a continuous amplitude function. Its saturation locus $\mathcal{L}_{\text{BI}} \subset \mathbb{R}$ is generically real-valued — its elements carry no information about shell structure.

The observable space decomposes discretely:

$$O = \bigsqcup_{n \in \mathbb{N}_q} O_n, \quad (15)$$

indexed by BFS shells of the Heisenberg Cayley graph [18]. No component $O_{n'}$ exists for non-admissible depths.

The proto-state is resolved at the first element of $\mathcal{L}_{\text{BI}} \cap \mathbb{N}_q$ — the first depth where Born–Infeld saturation coincides with an available shell. This is the Projection Locking theorem [18].

Theorem 5.12 (Discreteness from projection locking, after O22). *Under A3 and A4, quantum transitions are discrete. The discreteness is a consequence of the interaction of continuous Born–Infeld saturation with the discrete admissibility filter of Π_n , not a postulate.*

5.9 Three stable directions

The number of independent stable admissible directions in the fibre is exactly three, derived algebraically from the admissibility constraints alone [22]:

$$\begin{aligned} \text{BI parity} &\Rightarrow \text{fibre involution } c \leftrightarrow q - c \\ &\Rightarrow \text{non-abelian admissible support} \\ &\Rightarrow \mathbb{H}\text{-minimality} \\ &\Rightarrow \dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3 \\ &\Rightarrow \Sigma_c(n_3) = 3. \end{aligned} \tag{16}$$

The emergence of exactly three stable directions — and with it, three particle generations — is algebraic in origin, not geometric or phenomenological.

6 Entanglement and Bell Non-Applicability

6.1 Entanglement as shared ancestral fibre

Two observable systems A and B are entangled when they share a common ancestral fibre: a past state O_{n-k} in which they formed a single non-factorisable configuration.

The shared fibre F_{n-k} reduces the admissible directions for each subsystem: the admissible directions for A are constrained by B , and vice versa. The correlations arise not from any interaction at the moment of measurement, but from the persistence of the shared fibre — a residue of the past transition in which A and B were joined.

6.2 Why Bell’s factorisation assumptions do not apply

Bell’s theorem [30] constrains models in which measurement outcomes are determined by a variable λ satisfying: (i) λ is local (assignable separately to A and B), (ii) λ is present (defined at the moment of measurement), (iii) λ is independent (statistically independent of measurement settings).

The fibre variable in the present framework satisfies none of these conditions. It is not local (it belongs to the shared fibre, not to a spatial region), not present (it belongs to a past transition, not the current state), and not independent (A and B share the same ancestral fibre, which cannot be factorised).

The Bell factorisation assumption $P(a, b|\lambda) = P(a|\lambda) \cdot P(b|\lambda)$ fails because it requires λ to be assignable separately to A and B : the shared fibre is structurally non-factorisable. Bell-type inequalities are therefore not violated in this framework — their preconditions do not hold. This is established formally in the companion paper [1].

7 Summary: What Is Derived from What

Table 1 summarises the derivation chain.

Partial spectral stratification: an organising pattern from the companion programme

The four axioms A1–A4 are the axiomatic primitive of the programme. They do not by themselves determine the interaction content of the emergent theory; that content emerges from the spectral structure of the admissible operator, developed in the Q-series companion papers. The companion papers [10, 29, 26] establish the following partial pattern, which organises the physical interpretation of the Seeley–DeWitt expansion of $S_{\Pi}[g, A]$:

Principle 7.1 (Spectral stratification of effective interactions). *The physical role of an emergent interaction is determined not only by the symmetry it carries, but by the spectral order at which the admissible operator responds to variation. The gravitational sector appears at the first non-trivial horizontal response; the gauge sector appears at the vertical curvature response; and mixed gauge–gravity terms appear at the next coupled order.*

The evidence base for Principle 7.1 is as follows:

Spectral order	Sector	Geometric origin
a_2	gravity	horizontal variation of the base metric [10]
a_4	gauge	vertical variation of the fibre connection [29]
a_6	mixed coupling	joint base–fibre curvature terms [26]
a_{spin}	fermionic structure	Dirac-type admissible operator (open)

Remark 7.2. *Principle 7.1 is not derived from A1–A4 alone. Its correct epistemic status is: established by the companion programme, consistent with A1–A4, and subject to the conditionalities of each companion paper. In particular, the identification $G_{\Pi} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ remains conditional on [H-color] [27, 25], the Lorentzian continuation of the spectral functional remains an open step, and the fermionic entry is a structural prediction, not a derived result. The test of this prediction is whether admissibility forces a Clifford module and a $\mathbb{Z}_2$ -grading on the admissible operator.*

Open directions

Q2 — Born rule beyond SU(2). The Born rule $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi \rangle|^2$ is derived in [8] for the SU(2) parity sector. A first step beyond this sector is taken in [15], which derives the Born rule, singlet correlator, and Tsirelson bound for the co-admissible spin- $\frac{3}{2}$ sector on the binary icosahedral group $2I$, and identifies SU(2) as the unique stable sector under

admissibility refinement in the large- p limit. The full extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case and remains the primary open direction.

O31/O32 — Colour sector and recursive non-injectivity. The $SU(2)$ derivation proceeds via conjugate pairs $\{c, q-c\}$. For primes $q \equiv 1 \pmod{3}$, the Weil representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ admits colour triplets $\{c_1, c_2, c_3\}$ satisfying $c_1 + c_2 + c_3 \equiv 0 \pmod{q}$ and pairwise non-conjugacy. Under Hypothesis [H-color] (triplet co-admissibility, numerically supported in O32 [25]), O31 [27] derives $SU(3)$ as the admissibility fixed point of the colour fibre, completing the identification $G_{\text{SM}} = SU(3) \times SU(2) \times U(1)$.

Each element c_i of a valid triplet already carries spin- $\frac{1}{2}$ (from its conjugate pair $\{c_i, q - c_i\}$, O29 [19]) and one colour charge (from the triplet structure). This is the minimal quantum-number content of a quark. The identification of quarks as stable projected configurations in $\text{Im } \Pi$ carrying these quantum numbers, and the derivation of colour confinement as a structural consequence of the projection $\Pi_{\text{colour}} : \{c_i\} \rightarrow \{\text{colour-neutral objects}\}$ being non-injective (ENI [13], Corollary 6), are the primary open directions of the colour sector.

More broadly, ENI Corollary 6 establishes that the four axioms force non-injectivity not only at the level of the fundamental projection Π_1 , but at every stratum of a tower of genuinely distinct emergent levels. The programme’s current architecture instantiates two strata: the pair level (leptons, Π_1) and the triplet level (colour sector, Π_2). The structural derivation of the full particle hierarchy from this recursive tower is the central open direction of the programme.

Spectral quasi-fixed points and stable structures. The admissible neighbourhood $\mathcal{A}(O_n)$ varies generically with n (Remark 4.5). However, configurations O^* for which $\mathcal{A}(O^*)$ is approximately self-similar — in the sense that projection from O^* tends to return to a spectrally close neighbourhood — define *spectral quasi-fixed points* of the admissible flow. These are not static states, which would contradict the irreversibility established in Proposition 4.1, but projective orbits that traverse the same spectral region repeatedly. Stable physical structures — particles, gauge symmetries, and coupling constants — are candidates for identification with such quasi-fixed points: their observed stability would then be a projective property, arising from the near-invariance of their admissible neighbourhood under repeated projection, not from any ontological fixity. The universality of $a_q(s) \cdot q^2 \approx 2$ across all tested primes (O-series numerics) is consistent with the existence of such an attractor in the admissible spectral flow, independent of q . This numerical observation acquires a structural anchor from the companion paper

Q10 [24]: the horizontal coefficient $A_H(q)$ of the effective spatial co-metric satisfies $A_H(q) \rightarrow 2$ as $q \rightarrow \infty$, where the limiting value is identified as the Casimir eigenvalue

$$C_{\text{su}(2)}|_{j=1} = j(j+1) = 2 \quad (17)$$

of the spin-1 representation $\text{Sym}^2(V_\rho)$ — the unique $\text{su}(2)$ -invariant quadratic form on the effective spatial sector (Q10, Proposition 5.1, via Schur’s lemma). The epistemic chain leading to this identification is:

$$[X, Y] \neq 0 \Rightarrow \text{no perfect resolution} \Rightarrow [\text{U}] \Rightarrow \text{su}(2)\text{-isotropy} \Rightarrow C_{\text{su}(2)}|_{j=1} = 2 \Rightarrow A_H(q) \rightarrow 2, \quad (18)$$

where: the first implication is Corollary 4.6; the universality hypothesis [U] is structurally supported and numerically confirmed for $q \leq 211$ [21, 24]; the $\text{su}(2)$ -isotropy step is Q10 Lemma 4.1; and the Casimir identification is Q10 Proposition 5.1 (unconditional). The convergence $A_H(q) \rightarrow 2$ is established conditionally on [U] in Q10 Theorem 1.1. The attractor value 2 is therefore not a free parameter but is forced by the representation theory of the admissible fibre, providing a representation-theoretic interpretation of the quasi-fixed point. Formalising the existence and classification of quasi-fixed points of $\mathcal{A}$ is an open direction of the programme.

Theorem 5.7 establishes $F_n \simeq V_\rho$ at finite prime q . The relationship between this discrete structure and continuous spacetime geometry in the large- q limit — i.e. how V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$ approaches $L^2(\mathbb{R})$ and how the BFS shell structure produces a four-dimensional effective geometry — is addressed in the companion spectral geometry paper [11] (Section 3.5). That paper reduces Q5 to a single spectral tightness condition (Mosco tightness of the admissibility forms), confirms this condition quantitatively across all tested primes, and proves the embedding in the low-frequency regime relevant to admissible sequences. The identification of the effective metric from the principal symbol of the limit operator L_Π and the Lorentzian signature selection are established in the companion paper Q5b [2]. The temporal coefficient $A_\tau = 2$ is subsequently fixed by Q11 [28] via temporal Casimir rigidity, closing the effective co-metric to $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$.

Q14 — Fermionic structure and chirality. The spectral stratification of Principle 7.1 identifies the next open level of the programme: the fermionic sector. The companion papers Q12 and Q13 [29, 26] derive bosonic gauge and gravitational dynamics from a Laplace-type admissible operator $\mathcal{A}_{g,A}$. Fermionic structure requires a Dirac-type operator $\mathcal{D}_{g,A}$ satisfying $\mathcal{D}^2 = \mathcal{A}_{g,A} + \frac{1}{4}R$ (Lichnerowicz formula), whose $\mathbb{Z}_2$ -grading encodes chirality. The central open question is whether the admissibility structure forces a Clifford module structure on the admissible fibre and thereby selects a $\mathbb{Z}_2$ -grading: if so, the chiral asymmetry of the Standard Model becomes a structural consequence of admissibility; if not, this constitutes a structural boundary of the programme.

Concept	Source	Status
<i>Direct axiomatic consequences (A1–A4)</i>		
Irreversibility	Non-injectivity (A2)	Proved (Prop. 4.1)
Temporal order	Irreversibility + novelty (A2)	Proved (Thm. 4.2)
Proto-state (wave)	Unresolved multiplicity (A1, A3)	Derived (Prop. 5.2)
Phase coherence	Admissibility preserves open directions (A3)	Derived
Discrete transitions	Projection locking (A4, O22)	Proved (Thm. 5.12)
Three stable directions	$\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$ (O23)	Proved
Entanglement	Shared non-factorisable ancestral fibre	Derived
Bell non-applicability	Non-local, non-independent fibre	Proved [1]
Born rule (SU(2) sector)	Correlator + coherence [8]	Proved
$F_n \simeq V_\rho$ (Heisenberg structure)	Non-factorisability (A1, A3) + BI parity	Proved (Thm. 5.7)
<i>Key results of the companion programme (see [17] for the full derivation chain)</i>		
SU(2) as admissibility fixed point	Spectral convergence (Q2) [15]	Proved
Effective spacetime geometry, $D_{\text{hom}} = 4$; full co-metric $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$	BFS stratification (Q5b, Q6b, Q8, Q10, Q11) [2, 5, 20, 24, 28]	Proved
Einstein equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$	Spectral entropy variation [10]	Proved (structural)
SU(2) gauge structure	Quaternionic rigidity + Q6a [16, 9]	Proved
Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$	Spectral entropy a_4 variation (Q12) [29]	Proved (structural)
Gauge–matter coupling $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{YM}$	Joint variation $S_{\Pi}[g, A]$ (Q13) [26]	Proved (structural)
Hierarchy $G_N g_{YM}^2 \ll 1$ without fine-tuning	Seeley–DeWitt stratification (Q13) [26]	Structural
Tower of non-injectivity	ENI Corollary 6 [13]	Proved
<i>Conditional on named hypothesis</i>		
$G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$	Colour co-admissibility (O31, O32) [27]	Conditional on [H-color]
<i>Open</i>		
Full Hilbert structure (continuous spectrum)	Beyond discrete fibre; large- q limit	Open
Born rule (general observables)	Beyond SU(2)/ V_ρ sector	Open
Colour charges, quark sector	Individual sectors c_i (O31, O32)	Open

Table 1: Derivation chain from the four axioms. “Derived”: structural consequence of the axioms. “Proved”: explicit proof in this paper or a cited companion. “Proved (structural)”: established via structural argument internally consistent with the framework; full formalisation deferred. “Conditional”: proved subject to a named hypothesis.

8 Relation to Established Frameworks

The four axioms A1–A4 constitute a primitive layer from which geometry, dynamics, and quantum structure emerge without being presupposed. It is natural to ask how this primitive relates to the established formalisms of theoretical physics — not in the sense of reduction, but in the sense of structural translation: what each framework corresponds to, where it appears in the emergent hierarchy, and what it presupposes that the present framework derives.

Hamilton–Jacobi dynamics. Once the effective metric $g^{\mu\nu}$ has been reconstructed from the principal symbol of the admissibility operator (Q5b, Q6b [2, 5]), the characteristic propagation equation of the effective geometry coincides with the Hamilton–Jacobi equation for massless propagation. This identification belongs to the already projected geometric regime, downstream of the admissibility reconstruction. Hamilton–Jacobi theory should therefore be understood as an effective reformulation of the projected dynamics — valid once the effective configuration space has been constructed by the primitive admissibility relation — not as a primitive layer of the framework. The detailed identification, including the eikonal equation as principal-symbol characteristic, is given in Q6b [5].

Symplectic geometry and analytical mechanics. The framework of symplectic geometry takes a phase space — a cotangent bundle T^*M equipped with a canonical symplectic form — as its fundamental arena. In the present framework, this phase space is not a primitive: the effective configuration space M emerges from the Carnot–Carathéodory geometry of the BFS shell stratification (Q5b [2]), and the cotangent structure is induced by the principal symbol of the admissibility operator. Symplectic geometry is therefore the appropriate language for the effective dynamics on T^*M , valid after the configuration space has been constructed by the admissibility constraints; it does not describe the admissibility structure that produces it.

WKB approximation and geometric optics. In the WKB approximation, the quantum wave function takes the form $\psi \sim A e^{iS/\hbar}$, where the phase S is the Hamilton–Jacobi action. In the present framework, the phase of an admissible mode is not an input but an output of the fibre structure: it emerges from the coherence conditions imposed by the admissibility filter on the Weil representation [8]. The WKB identification therefore holds at the level of the effective description, once the continuum limit has been taken (Q5a [11]): the collective phase of the admissible modes plays the role of S , and geometric optics is the ray approximation of the admissible propagation structure in this limit. The WKB phase is thus a derived quantity,

encoding the projective compression of the admissible fibre, rather than a fundamental dynamical object.

Functional renormalisation group. The Wetterich effective average action satisfies an exact renormalisation group equation whose infrared fixed point recovers the physical effective action. In the present framework, the Mosco limit of the admissibility Dirichlet forms (Q5a [11]) plays a structurally analogous role: the sequence $\{\mathcal{E}_q\}$ flows, as $q \rightarrow \infty$, to the scalar operator $\mathcal{L}_\Pi$, whose principal symbol defines the effective metric. The spectral entropy functional $S_\Pi[g, A]$ of the companion programme is the effective action resulting from this limit [10, 29, 26]. The structural difference is that the Wetterich flow renormalises a pre-existing field theory, while the present limit produces the field content itself from the admissibility constraints.

Page–Wootters construction and relational time. The Page–Wootters construction derives apparent temporal evolution from conditional states of a globally stationary wave function, treating time as a relational label on a clock subsystem. The projective temporal ordering of the present framework [14] shares this relational spirit — time is not a parameter of the substrate but a monotone ordering induced by the admissibility cascade — but differs in one structural respect. In the Page–Wootters model, the global state is fully recoverable (the projection is effectively injective), whereas here the projection Π is structurally non-injective (A2, ENI [13]). Temporal ordering in the present framework is therefore irreversible at the structural level: it reflects the irreversible accumulation of projected spectral capacity, not the choice of a clock observable.

Holographic renormalisation and radial Hamilton–Jacobi. In holographic renormalisation, the on-shell gravitational action satisfies a radial Hamilton–Jacobi equation whose flow encodes the renormalisation group of the boundary field theory. The Schwarzschild geometry derived in Q6b [5] from flux conservation of the admissibility operator has a structural parallel with this picture: both frameworks derive a bulk geometry from a propagation constraint, and both identify the horizon as a degeneration of the effective description. The structural difference is that holographic renormalisation presupposes a fixed anti-de Sitter background and a specific bulk-boundary duality, while the present framework derives the effective geometry from the admissibility operator without a background or a boundary field theory.

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